R_A at ArA-DF 2026: Fine-Tuned Self-Supervised Encoders with Acoustic Degradation For Robust Arabic Deepfake Detection

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Abstract

We describe our submission to ArA-DF 2026, the ArabicNLP shared task on Arabic speech deepfake detection, in which systems classify audio as real or deepfake. We participated in both tracks: Track 1 (Dialect Generalization) and Track 2 (Acoustic Robustness). Our systems investigated finetuning three large self-supervised speech encoders — WavLM, wav2vec2-XLS-R, and HuBERT — with a learnable attentive-pooling head. We evaluated each encoder separately and further combined encoders through a logistic-regression meta-classifier. In addition to improve robustness we apply an acoustic degradation pipeline (additive noise, lossy codec re-encoding, band-limiting, and gain) to 50% of the training data. For track one the best achieved result is, a fully fine-tuned WavLM with attentive pooling and augmentation, which scored 2.19% EER test set, and a deepfake-pretrained XLS-R achieves 7.80% EER on Track 2 — which out performed the official baseline. Our ablation study shows that augmentation and backbone choice are the primary factors, while adding correlated ensemble members yields no gain.

1 Introduction

Speech synthesis and voice-conversion systems have advanced to the point where synthetic speech is difficult to distinguish from genuine recordings by ear: in a controlled listening study, participants identified deepfake audio correctly only 73% of the time, and familiarising them with examples improved performance only marginally (Mai et al., 2023). Reliable automatic detection is therefore an increasingly important safeguard against fraud, impersonation, and misinformation, motivating benchmark efforts such as the ASVspoof series, whose 2021 edition introduced a dedicated speech deepfake track (Yamagishi et al., 2021; Liu et al., 2023).

Recent work replaces hand-crafted features with self-supervised speech representations. Encoders such as wav2vec2.0 (Baevski et al., 2020), its multilingual extension XLS-R (Babu et al., 2022), HuBERT (Hsu et al., 2021), and WavLM (Chen et al., 2022) transfer strongly to anti-spoofing, where fine-tuning an SSL backbone and extracting a compact audio embedding have now become state-of-the-art approaches (Tak et al., 2022).

Although some works have proposed solutions for Arabic deepfake detection (Issa, 2026; Elsetohy et al., 2026), the field remains under-explored, especially when handling out-of-domain or distorted recordings.

ArA-DF 2026 provides a focused benchmark for detecting deepfake Arabic audio from genuine recordings. Each recording is labelled real, for genuine human speech, or spoofed, for synthetic speech produced by text-to-speech or voice-conversion systems, and systems are ranked by Equal Error Rate (EER), a threshold-independent metric standard in anti-spoofing evaluation. The two tracks probe complementary failure modes. Track 1 (Dialect Generalization) measures whether a detector remains reliable on dialects and speakers absent from training. Track 2 (Acoustic Robustness) tests resilience to practical distortions such as compression artefacts, background noise, and re-recording effects.

2 Task Setup and Data

The ArA-DF 2026 dataset is released by ArabicSpeech. As shown in Table 1, it consists of 22,500 labelled training and 21,000 labelled development recordings, with unlabelled development-test and blind final-test sets for each track. The Track 1 final-test set contains 144,210 recordings and the Track 2 final-test set 127,746.

The organisers provide a baseline pairing an XLS-R 300M front-end with an AASIST back-end (Jung et al., 2022), reporting 14.54% EER on the Track 1 final test and 27.11% on Track 2.

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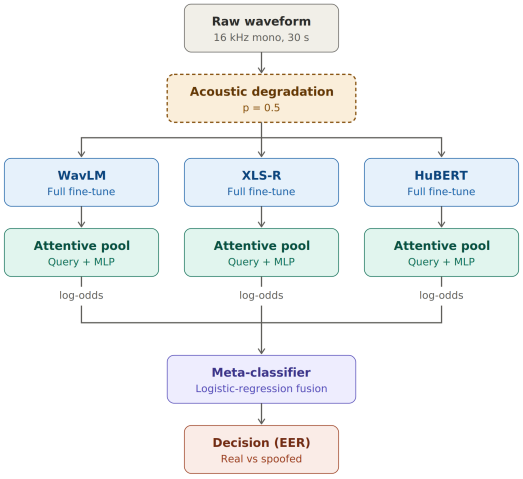


Figure 1: Overview of the proposed system. The raw waveform is first passed through the acoustic degradation pipeline, drawn with a dashed outline to indicate that it is applied stochastically with probability $p = 0.5$. The (optionally degraded) waveform then feeds three fine-tuned self-supervised encoders, each with an attentive-pooling head. The resulting bonafide log-odds are fused by a logistic-regression meta-classifier to produce the real-vs-spoofed decision.

Split	Rows	With Labels
Train	22,500	Yes
Dev	21,000	Yes
Track 1 development-test	16,023	No
Track 1 final-test	144,210	No
Track 2 development-test	14,193	No
Track 2 final-test	127,746	No

Table 1: Dataset splits and label availability. Official scoring uses the blind final-test sets.

Systems are ranked by **Equal Error Rate (EER)**, the standard anti-spoofing metric. A detector outputs a score per utterance; applying a threshold yields a false-acceptance rate (spoofed audio accepted as genuine) and a false-rejection rate (genuine audio rejected). As the threshold varies, one rises while the other falls, and the EER is the operating point where they are equal. Lower is better, and because it is read at this fixed crossover, EER summarises performance without committing to a threshold.

3 System Overview

The following subsections describe our system: the overall architecture (Section 3.1), the self-supervised encoders and pooling head that turn an utterance into a score (Sections 3.1.1–3.1.2), the fine-tuning regimes we compared (Section 3.2), the acoustic augmentation pipeline (Section 3.3), and the meta-classifier that fuses encoders (Section 3.4).

3.1 Proposed Architecture

All of our detectors share a common template: a pre-trained SSL Transformer encoder maps the raw waveform to a sequence of frame embeddings, which a pooling head aggregates into a single vector for binary classification.

3.1.1 Self-supervised encoders

We use three encoders at this stage:

- **WavLM-Large** (Chen et al., 2022), pre-trained predominantly on English speech with joint masked-speech prediction and denoising, which suits noisy conditions.
- **XLS-R** (Babu et al., 2022), a wav2vec2.0 model (Baevski et al., 2020) pre-trained on 436k hours across 128 languages; we adopt a publicly available variant already fine-tuned for deepfake audio classification.
- **HuBERT-Large** (Hsu et al., 2021), pre-trained on English speech, supplying a third, independently pre-trained representation.

Of the three encoders, only XLS-R has substantial multilingual exposure, yet we use all three despite their mostly non-Arabic pre-training. Spoofing detection relies less on linguistic content than most speech tasks: the key cues are low-level synthesis artefacts — unnatural spectra, phase inconsistencies, and vocoder signatures — that are largely language-independent, and SSL encoders fine-tuned for anti-spoofing transfer well across languages (Tak et al., 2022). Language still has some effect, since dialectal acoustics change the genuine-speech distribution, but much less than in recognition tasks; in fact, English-pretrained WavLM gives our strongest Track 1 result after fine-tuning.

3.1.2 Pooling heads

We compare two pooling strategies.

Mean pooling averages frame embeddings over valid positions and passes the result through an MLP.

Attentive pooling introduces a single learnable query vector $\mathbf{q}$ that attends over the sequence of frame embeddings $\mathbf{H}$ via multi-head attention, $\mathbf{u} = \mathrm{MHA}(\mathbf{q}, \mathbf{H}, \mathbf{H})$, with padded positions masked out. The query learns to assign higher attention weights to frames that are more useful for the real-vs-spoofed decision, producing a weighted sum

rather than the uniform average of mean pooling. The pooled vector $\mathbf{u}$ is layer-normalised and passed through an MLP.

3.2 Fine-tuning strategies

We contrast three fine-tuning methods: (i) **head-only** training with a frozen encoder; (ii) **full fine-tuning**, updating the entire encoder and head jointly from the first step; Full fine-tuning gave the best results, but requires care: a randomly initialised head can destabilise pre-trained weights early in training, so we use a low encoder learning rate, linear warmup, and gradient clipping.

3.3 Data Degradation

To improve robustness, we degrade 50% of the training records. This forces the model to focus on the underlying synthesis features that persist through degradation, rather than on shallow cues tied to clean-recording conditions that fail to generalize to real-world degradations (codec compression, background noise, and social-media re-encoding).

Each degraded clip passes through a random chain of the following:

- **Additive Gaussian noise** at 8–30 dB SNR;
- **Lossy codec re-encoding** (MP3 at 32–96 kbps, or AAC, Opus, or Ogg Vorbis at low bitrates), emulating social-media transmission;
- **Band-limiting** with a high-pass filter at 50–150 Hz and a low-pass filter at 6–8 kHz, approximating telephone-quality channels;
- **Random gain shift** of ± 6 dB.

3.4 Meta-classifier

Using the validation set, we selected the best-performing backbones based on their individual classification performance and used them to train a meta-classifier. The selected backbones were kept frozen, and their bonafide log-odds were combined and fed into a logistic-regression meta-classifier, whose learned weights were then applied to test and blind-evaluation scores.

4 Experimental Setup

Data usage. Because final-test labels are withheld, model selection relies on an internal split. We merge the labelled train and dev pools, hold out 10% as an internal test set and a further 10% of the remainder as validation, with a fixed seed so every experiment sees identical partitions. Validation is used for early stopping and for fitting the meta-classifier; the internal test set is used for ablations. Development-test and final-test EERs are official leaderboard scores.

Hyperparameter	Value
Sample rate	16 kHz
Max audio length	30 s
Encoder LR (full FT)	1×10^{-5}
Head LR	1×10^{-4}
Optimiser	AdamW, wd 1×10^{-4}
LR schedule	cosine, 5% warmup
Effective batch size	$\approx$32
Gradient clipping	1.0
Augmented fraction	0.5
Noise SNR range	8–30 dB

Table 2: Principal training and augmentation hyperparameters, shared across full-fine-tuning experiments.

Preprocessing. Audio is decoded from FLAC, converted to mono, resampled to 16 kHz, and truncated to 30 seconds; waveforms are normalised by the encoder's feature extractor. Class imbalance is handled with inverse-frequency weighting in the loss.

Hyperparameters. Table 2 lists the principal settings. Separate learning rates for the pre-trained encoder and the randomly initialised head prevent catastrophic forgetting while letting the head adapt quickly.

Metric. EER is the operating point where false-acceptance and false-rejection rates coincide. It is symmetric under label inversion, so the choice of positive class does not affect its value; we verified our internal EERs matched the official development-test leaderboard before the blind evaluation.

5 Results and Analysis

In the following subsections, we will present the results for the development set and the testing set.

5.1 Development Set Results

Table 3 shows the best results for the development set. The fully fine-tuned WavLM with attentive pooling and 50% augmentation (Exp 1) is our strongest individual model on Track 1, reaching **2.30% EER**. For Track 2, the deepfake-pretrained XLS-R with 50% degradation (Exp 2) is best at **7.80% EER**.

Exp ID	Encoder	FT	Degradation	T1-dev	T2-dev
1	WavLM	Full	50%	**2.30**	16.73
2	XLS-R	Full	50%	3.41	**7.80**
3	WavLM	Full	—	6.30	9.66
4	WavLM‡	Head	—	6.10	19.95
5	HuBERT‡	Head	—	7.83	21.28
6	XLS-R	Full	—	8.16	—

Table 3: Individual system EER (%) on the official development-test split; best per column in **bold**. All systems use attentive pooling except those marked ‡ (mean pooling).

Three ablation patterns are notable. First, **fine-tuning depth matters**: full fine-tuning beats head-only training for every encoder. The two head-only systems (Exps 4 and 5) are the weakest on Track 1. Second, **attentive pooling beats mean pooling** across every encoder and fine-tuning regime we tested. Third, **the effect of degradation is track- and backbone-dependent**: adding 50% degradation to WavLM lowers Track 1 development EER from 6.30% to 2.30% (Exps 3 vs. 1), and the XLS-R system (Exp 2) far outperforms its un-degraded counterpart (Exp 6) on Track 2, where the latter fails to produce a usable result. Degradation is not uniformly beneficial, however: for WavLM on Track 2 it hurts, raising development EER from 9.66% (Exp 3) to 16.73% (Exp 1). The gains from degradation are thus clearest on Track 1 and for the deepfake-pretrained XLS-R on Track 2; the latter suggests that an un-degraded XLS-R detector keys on channel characteristics that the Track 2 distortions destroy, while WavLM's un-degraded Track 2 behaviour does not follow the same pattern.

5.2 Official Testing Set Results

We took the best two models for each track and evaluated them on the official final test set, shown in Table 4.

Exp ID	Encoder	FT	Degradation	T1-test	T2-test
1	WavLM	Full	50%	**2.1948**	—
2	XLS-R	Full	50%	3.4112	**7.7960**
3	WavLM	Full	No	—	9.7202

Table 4: Individual system EER (%) on the official final test set; best per column in **bold**.

We further combined the best two models for each track using a logistic-regression meta-classifier, reported in Table 5.

5.3 Discussion

Our best systems are strikingly stable between the development and blind final-test splits. On Track 1, the augmented WavLM (Exp 1) moves from 2.30% to **2.19%** EER, and on Track 2 the augmented XLS-R (Exp 2) holds at **7.80%**, essentially unchanged despite the far larger final-test sets — indicating that model selection did not overfit the development split.

System	T1-Test	T2-Test
META (1+2)	**2.39**	—
META (2+3)	—	**7.21**

Table 5: Meta-classifier and multi-layer aggregation, EER (%). Systems are indexed by the constituent model IDs from Table 4.

Two factors drive these results: First, **degradation helps the XLS-R Track 2 system**: on Track 2 the augmented XLS-R (7.80%) far outperforms its un-degraded counterpart, and the un-degraded WavLM (Exp 3) trails at 9.72%, indicating that for XLS-R, without exposure to codec, noise, and band-limiting artefacts a detector keys on channel cues rather than synthesis artefacts. The benefit is not universal — degradation hurts WavLM on Track 2 (Table 3) — so it is best read as a backbone-specific effect rather than a blanket robustness gain. Second, **fusion rewards decorrelation**: pairing the augmented XLS-R with the un-degraded WavLM (META 2+3) improves Track 2 to **7.21%**, below either constituent, because the two make complementary errors.

6 Conclusion

In this paper we presented our participation in both tracks in ArA-DF 2026. We proposed an approach that employ self-supervised finetuned models for Arabic speech deepfake detection. Full fine-tuning of WavLM and XLS-R with attentive pooling plus an acoustic degradation pipeline achieves 2.19% EER on the Track 1 final test and 7.80% on Track 2, which greatly outperformed the AASIST baseline. Future works include other attention mechanisms for frame fusion, wider range of degradation styles, and per-dialect error analysis.

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